RF Signal Strength Prediction & IRS Optimization

ACADEMIC PROJECT SUMMARY & MILESTONE REPORT

1. Project Overview

This report details the systematic development of an intelligent framework designed to predict and optimize Radio Frequency (RF) signals using Intelligent Reflecting Surfaces (IRS). The project progressed through discrete developmental phases, evolving from a passive Machine Learning (ML) prediction framework into a robust, spatially aware 3D model, and ultimately culminating in an active, closed-loop optimization system.

2. Phase 1: Baseline Machine Learning Models

In the initial phase, a synthetic RF dataset was generated to simulate fundamental signal propagation mechanics. Four distinct ML algorithms were trained to predict the Received Signal Strength Indicator (RSSI). The models evaluated the complex, non-linear degradation of signals based on generalized distance assumptions, frequency, and environmental factors. **Gradient Boosting** emerged as the most accurate predictive model.

ML Algorithm	RMSE (Lower is Better)	R^2 Score (Higher is Better)
Gradient Boosting	**4.9555**	**0.9230**
Linear Regression	6.0833	0.8840
Random Forest	6.3595	0.8732
Decision Tree	8.7148	0.7619

3. Phase 2: 3D Coordinate Integration

To drastically improve simulation realism, the dataset logic was upgraded to include exact 3D spatial coordinates (X, Y, Z axes). Precise geometric locations were assigned to the Transmitter (Tx), the IRS panel, and the user Receiver (Rx). This allowed the system to compute exact Euclidean path distances rather than relying on randomized assumptions. This spatial awareness yielded notable improvements in absolute error (RMSE) across all predictive models.

ML Algorithm	Original RMSE	3D Spatial RMSE	3D Spatial R^2
Gradient Boosting	4.9555	**4.9143**	0.8947
Linear Regression	6.0833	**5.0894**	0.8871
Random Forest	6.3595	**6.0400**	0.8410

4. Phase 3: IRS Angle Optimization (Genetic Algorithm)

The final phase marked a critical paradigm shift in the project's utility: transitioning from passive signal "Prediction" to active signal "Control". Rather than merely estimating signal degradation, the system was re-engineered to intervene. A **Differential Evolution (Genetic Algorithm)** was implemented as an objective optimization function. By taking real-time 3D user coordinates as fixed parameters, the algorithm actively searches through thousands of iterations to find the optimal IRS reflection angle (phase shift pattern) that mathematically maximizes the RSSI for that specific user.

5. Final Experimental Results

A live simulation was executed targeting a specific user positioned at a distance of 1,618 meters in a Suburban environment. The Genetic Algorithm successfully deduced the optimal reflection pattern, yielding a significant increase in signal strength.

ORIGINAL RSSI	OPTIMAL IRS ANGLE	OPTIMIZED RSSI	TOTAL IMPROVEMENT
-74.43 dBm	**89.17°**	**-59.84** dBm	**+14.59** dB
Pre-optimization	Deduced Pattern	Post-optimization	Massive Net Gain

6. Conclusion

> The project successfully evolved from a theoretical Machine Learning prediction pipeline into a practical, closed-loop intelligent control system. By combining highly accurate 3D spatial modeling with advanced genetic optimization algorithms, the system demonstrates the ability to actively correct and enhance radio frequencies in real-time. The achieved gain of **14.59 dB** proves the immense viability of IRS-assisted networks, providing a robust software foundation for next-generation 6G telecommunications.